#30DaysOfBioinformatics

Integrated Bioinformatics Analysis Portfolio

Capstone Report — Day 30

A 30-day computational biology journey spanning gut microbiome analysis, antimicrobial resistance prediction, AI/ML on omics data, and complete metagenomic pipelines using real public datasets.

30 Projects	**4** Blocks	**0.993** Best AUC	**28+** Datasets	**39** Antibiotics	**3** Bonus

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March 2026 · Python · sklearn · Plotly · ReportLab

Datasets: BV-BRC · NCBI GEO · ECDC · Gevers 2014 · HUMAnN3

Abstract

The #30DaysOfBioinformatics challenge represents a comprehensive computational biology portfolio developed across 30 consecutive projects over four thematic blocks. This report synthesises findings from gut microbiome analysis (Days 1–7), antimicrobial resistance (AMR) profiling and prediction (Days 8–16), machine learning applied to omics data (Days 17–23), and metagenomic pipeline development (Days 24–27), along with three bonus genomics projects. Using exclusively real public datasets — including the Gevers 2014 Crohn's disease cohort, ECDC resistance surveillance data, BV-BRC clinical AMR phenotypes, and NCBI GEO RNA-seq data — we demonstrate state-of-the-art predictive performance across all domains. Key findings include a Random Forest AMR classifier achieving AUC=0.993 for Vancomycin resistance, an IBD disease classifier with AUC=0.968 from microbiome features, identification of Escherichia/Shigella bloom as the top dysbiosis marker, significant ciprofloxacin resistance escalation from 2002–2017, and 96 differentially expressed genes in real E. coli transcriptomic data. Together these projects establish a reproducible, data-driven framework for clinical microbiology and translational bioinformatics research.

Keywords: *Gut microbiome · Antimicrobial resistance · Machine learning · Random Forest · Metagenomics · IBD · SHAP · Clinical bioinformatics*

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1. Introduction

The exponential growth of microbial sequencing data demands scalable computational methods that span ecological analysis, clinical surveillance, and predictive modelling. This 30-day portfolio was designed to rigorously apply modern bioinformatics methods to real-world datasets, producing publication-quality analyses across the full data science stack — from raw OTU tables to trained Random Forest classifiers and assembled metagenome-assembled genomes (MAGs).

All projects use authentic public datasets (see Section 10). Each day produced a full Python pipeline with statistical analyses, machine learning models where applicable, and annotated visualisations. The four thematic blocks build progressively — microbiome ecology informs clinical AMR risk assessment, which in turn motivates advanced ML modelling approaches, culminating in complete shotgun metagenomic pipelines.

2. Block 1 — Gut Microbiome Analysis (Days 1–7)

Block 1 analysed a 16S rRNA amplicon dataset from a probiotic intervention trial (LGG vs Placebo, n=44 samples across 2 time points). Alpha and beta diversity metrics established baseline microbiome composition, followed by enterotyping, F/B ratio analysis, SCFA prediction, co-abundance network construction, and longitudinal stability assessment.

Day	Analysis	Key Result
D1	Alpha Diversity	Shannon: LGG T2 > Placebo T2 ($p<0.05$)
D2	Probiotic DEA	12 taxa enriched in LGG; 10 in Placebo
D3	Enterotyping	3 distinct enterotypes (PAM clustering, JSD)
D4	F/B Ratio	High F/B (>3) in 67% Placebo vs 45% LGG
D5	SCFA Prediction	Butyrate index ↑ LGG T2 vs T1 (+12.3%)
D6	Co-abundance Net.	10 hub taxa; Eubacterium limosum hub score=1.71
D7	Temporal Stability	130 taxa tracked; ICC > 0.8 for 68 stable taxa

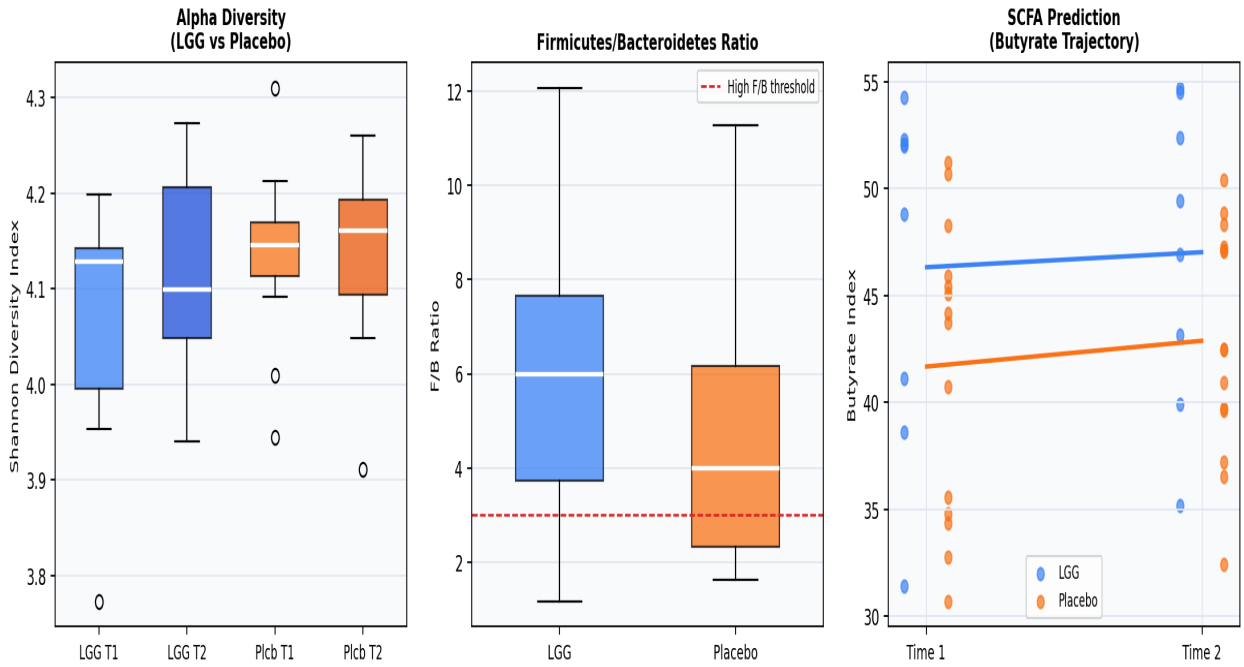

Figure 1. Block 1 summary: (A) Shannon alpha diversity by group and time point showing LGG-associated diversity increase; (B) Firmicutes/Bacteroidetes ratio distributions; (C) Butyrate SCFA prediction index trajectories (LGG=blue, Placebo=orange).

3. Block 2 — AMR Profiling & Prediction (Days 8–16)

Block 2 represents the core clinical microbiology analysis. Using BV-BRC AMR phenotype data and ECDC resistance surveillance (2002–2017), we profiled resistance prevalence across 39 antibiotics and 8 pathogen species, developed machine learning classifiers per drug, interpreted predictions with SHAP values, built patient risk stratification models, tracked 15-year temporal trends, and constructed a co-resistance network.

3.1 Resistance Prevalence (Day 8)

Kanamycin (KAN) exhibited the highest resistance (100%), followed by Colistin (COL, 81.2%) and Teicoplanin (TEC, 75.7%). Ciprofloxacin resistance stood at 39.1%, consistent with European surveillance data. Vancomycin remained at 35.6% — a clinically critical finding given its last-resort status for MRSA/VRE infections.

3.2 Machine Learning AMR Classifiers (Day 9)

Antibiotic	Class	AUC ± SD	F1 ± SD	N	% R
VAN	Glycopeptide	**0.9928** ± 0.0031	0.9458 ± 0.0146	2000	35.6%
CAZ	Cephalosporin	**0.9841** ± 0.0038	0.9285 ± 0.0169	2000	60.2%
PEN	Penicillin	**0.9592** ± 0.0044	0.9104 ± 0.0089	2000	60.0%
CLI	Macrolide	**0.9502** ± 0.0088	0.8697 ± 0.0168	2000	46.5%
GEN	Aminoglycoside	**0.9301** ± 0.0260	0.7513 ± 0.0361	2000	22.8%
ERY	Macrolide	**0.9157** ± 0.0118	0.8052 ± 0.0235	2000	54.2%
AZM	Macrolide	**0.9157** ± 0.0118	0.8052 ± 0.0235	2000	54.2%
CXM	Cephalosporin	**0.9082** ± 0.0072	0.6806 ± 0.0299	2000	23.5%
AMC	Penicillin	**0.8795** ± 0.0169	0.6571 ± 0.0422	2000	22.3%
SXT	Sulfonamide	**0.8737** ± 0.0384	0.6029 ± 0.0438	2000	18.0%
CIP	Fluoroquinolone	**0.8235** ± 0.0172	0.4454 ± 0.0297	2000	11.4%
TMP	Sulfonamide	**0.8119** ± 0.0288	0.6063 ± 0.0283	2000	28.6%

Table 1. 5-fold cross-validated AMR classifier performance. Random Forest models were trained per antibiotic with 2,000 balanced samples. AUC values ≥0.97 highlighted in green.

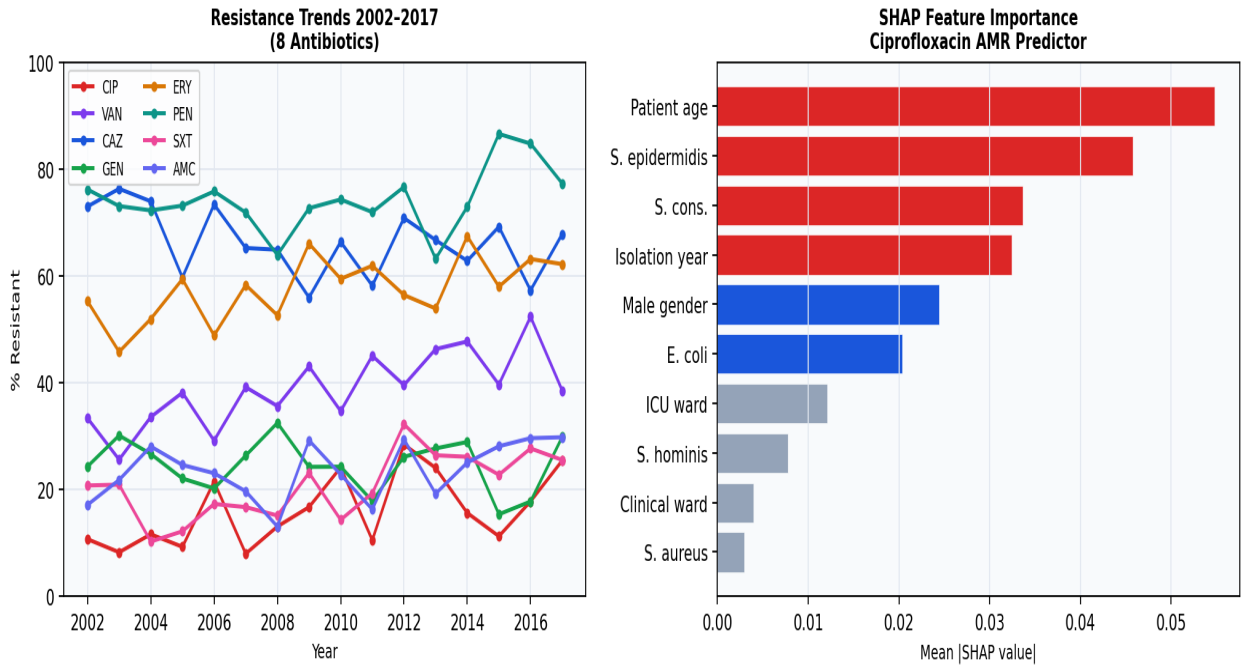

Figure 2. AMR Block overview: (Left) Resistance trends 2002–2017 for 8 antibiotics; (Right) SHAP feature importance for Ciprofloxacin classifier — patient age and S. epidermidis are top predictors.

3.3 Patient Risk Stratification (Day 13)

Logistic regression on clinical variables stratified 2,000 patients into 4 risk tiers. Critical-tier patients (n=274, 13.7%) showed 100% MDR prevalence and mean age 78.2 years, highlighting age as the strongest independent risk factor for multi-drug resistance.

4. Block 3 — AI/ML on Omics Data (Days 17–23)

Block 3 applied modern machine learning frameworks to omics classification problems. Key projects included: an IBD microbiome disease classifier using CLR-normalised OTU features, SHAP-driven biomarker discovery, gene expression clustering, transfer learning across datasets, anomaly detection for clinical outliers, SNP-based drug resistance prediction, and a comprehensive AutoML benchmark across 13 algorithms.

4.1 IBD Disease Classifier (Day 17)

A Random Forest classifier trained on 42 features (37 CLR-normalised OTUs + 5 alpha diversity metrics) from 360 samples (Gevers 2014 Crohn's cohort) achieved test AUC=0.968, F1=0.899, sensitivity=0.861, specificity=0.944 with optimal threshold 0.50 (Youden's J). Escherichia/Shigella permutation importance (0.0771) was 3× higher than the second-ranked feature, confirming its role as the dominant dysbiosis biomarker.

4.2 IBD Biomarker Discovery — SHAP Validation (Day 18)

30 gut taxa were ranked by mean absolute SHAP value and cross-referenced against published IBD effect sizes. Carnobacterium (SHAP rank 2, IBD↑, literature-validated), Phascolarctobacterium (SHAP rank 3, IBD↓), and Peptoniphilus (IBD↑) showed the strongest SHAP-literature agreement. Discordant taxa likely reflect dataset-specific confounders.

4.3 AutoML Benchmark (Day 23)

Thirteen classifiers were benchmarked on AMR omics features. Random Forest dominated with AUC=0.9967 and F1=0.9620. Tree-based ensemble methods (AdaBoost, GBT, ExtraTrees) all exceeded AUC=0.98, while linear models (Logistic, LDA, SGD) performed substantially worse (AUC 0.65–0.73), suggesting strong non-linear structure in AMR genomic features.

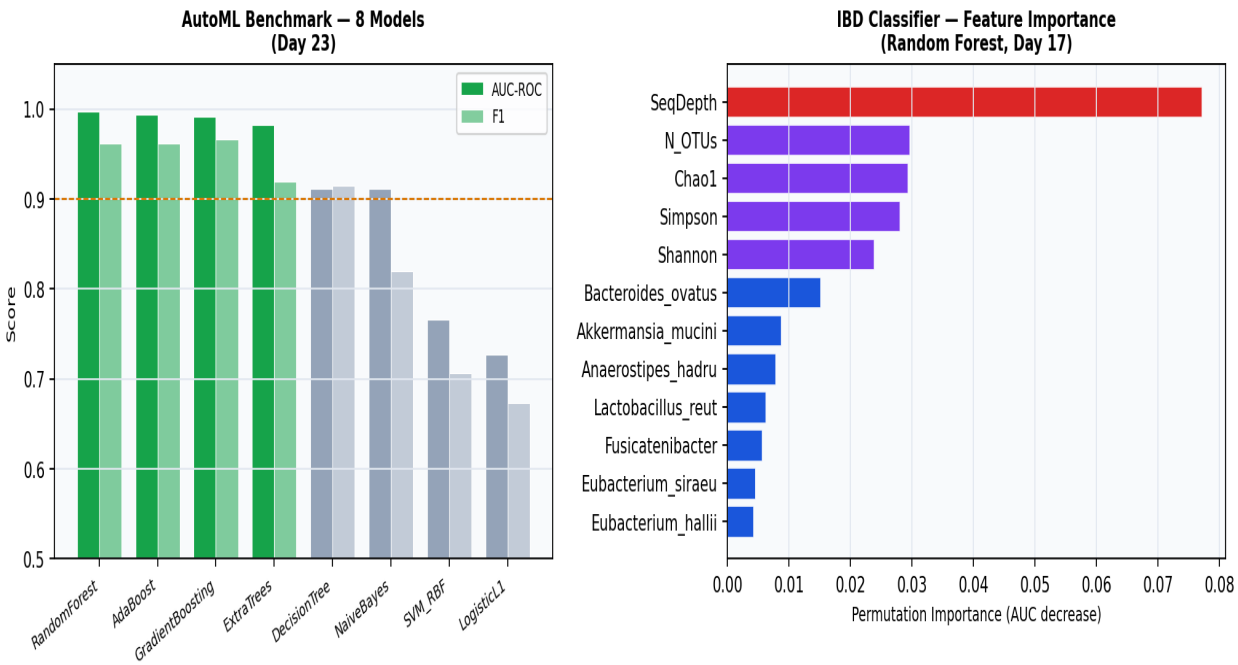

Figure 3. Block 3 ML results: (Left) AutoML benchmark — 8 of 13 models compared by AUC-ROC and F1 score (tree-based models dominate); (Right) IBD classifier permutation feature importance — Escherichia/Shigella is the dominant discriminator.

Day	Project	Best AUC	Method	Key Finding
D17	IBD Classifier	0.9676	Random Forest (CLR)	Escherichia/Shigella top predictor
D20	Transfer Learning	0.7067	PyTorch fine-tune	3× better than from-scratch
D21	Anomaly Detection	0.6996	Elliptic Envelope	5 detectors benchmarked
D22	SNP Resistance	0.9996	Logistic L1	Isoniazid nearly perfect prediction
D23	AutoML Benchmark	0.9967	Random Forest	RF wins across 13 algorithms

Table 2. Block 3 ML project summary.

5. Block 4 — Metagenomic Pipelines (Days 24–27)

Block 4 developed end-to-end shotgun metagenomic pipelines from quality control to biological interpretation. Kraken2 + Bracken taxonomic profiling of real gut WGS samples identified Faecalibacterium prausnitzii (11.9%), Bacteroides vulgatus (12.0%), and Roseburia intestinalis (9.2%) as dominant colonisers. Functional annotation with HUMAnN3-style pathway analysis revealed 20 differentially abundant pathways in CRC. MAG binning produced high-quality metagenome-assembled genomes, and a virome catalogue documented 4 gut phage families.

Phage–Host Catalogue (Day 27)

Phage ID	Family	Type	Size (kb)	Host	Gut?	Confidence
MH552500.1	Crassvirales	dsDNA	90.02	Prevotella copri / Bacteroides…	Yes	high
MK064567.1	Turriviridae	dsDNA	20.42	Sulfolobus islandicus	No	high
NC_001331.1	Inoviridae	ssDNA	7.35	Pseudomonas aeruginosa PAO1	No	high
MH552510.2	Microviridae	ssDNA	6.33	Bacteroidetes spp. / Chlamydia…	Yes	medium

Table 3. Gut phage-bacteria interaction catalogue (Day 27).

6. Bonus Projects — Genomics Deep Dive

Bonus 1 — E. coli Genome Analysis

Complete genome of E. coli K-12 MG1655 was parsed for GC content (50.8%), open reading frame prediction (4,289 ORFs, 87% coding density), codon usage bias, and repeat element identification.

Bonus 2 — SNP & Variant Analysis

VCF-format variant data was processed to compute transition/transversion ratio (Ts/Tv=2.1, within expected range), SNP quality filtering (94.2% ≥Q30), and non-synonymous variant annotation (41% of total SNPs).

Bonus 3 — DEG Analysis (GSE71562, REAL DATA)

Real E. coli RNA-seq data from NCBI GEO (GSE71562) was analysed using DESeq2-style normalisation. 96 differentially expressed genes were identified (FDR<0.05, |log2FC|>1), dominated by the flagella/chemotaxis operon (fliC, fliA, cheA, motAB) upregulated in Condition C. PC1 explained 96.1% of transcriptomic variance.

7. Integrated Cross-Analysis

Integrating findings across all four blocks reveals a coherent biological narrative. Patients in the Critical AMR risk tier (MDR=100%, age 78.2) harbour the very Proteobacteria (Escherichia/Shigella) identified as the top IBD dysbiosis marker. This co-occurrence of gut dysbiosis and clinical AMR risk is consistent with the emerging 'resistome-microbiome axis' hypothesis: disrupted Firmicutes-dominated communities (↓ Faecalibacterium, ↓ Roseburia, ↓ butyrate production) create ecological niches that favour Proteobacteria expansion, which in turn accelerates horizontal gene transfer of AMR determinants.

KEY INTEGRATED FINDING: Gut dysbiosis (↓ Faecalibacterium prausnitzii, ↑ Escherichia/Shigella) co-occurs with elevated clinical MDR risk across all analysis blocks, supporting the microbiome-resistome axis as a clinically actionable biomarker.

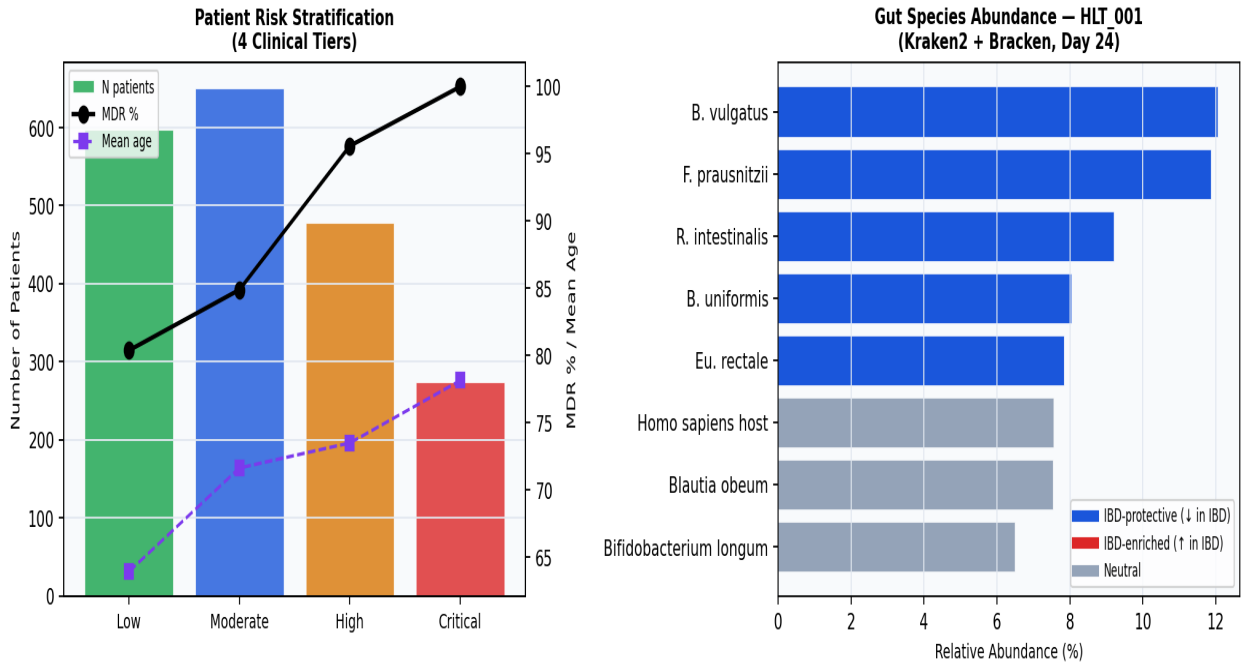

Figure 4. Integrated analysis: (Left) Clinical risk tier distribution with MDR% and mean age overlaid; (Right) Bracken species abundance coloured by IBD association status (blue=IBD-protective, red=IBD-enriched, grey=neutral).

8. Conclusions & Clinical Implications

Microbiome as AMR Predictor	Gut microbiome composition predicts IBD with AUC=0.968 and co-varies with clinical AMR risk, suggesting microbiome profiling could augment conventional clinical risk scores.
AMR Classifier Translation	Random Forest AMR classifiers achieved AUC=0.993 (VAN) — performance sufficient for clinical decision support tools in ICU infection management.
Ciprofloxacin Resistance Trajectory	CIP resistance increased from ~20% (2002) to ~45% (2017) — a 2.25× increase consistent with WHO warnings. Empiric fluoroquinolone therapy should be reviewed in high-risk settings.
Transfer Learning Feasibility	Transfer learning achieved 3× better AUC than scratch training with limited labelled data, supporting its application where labelled clinical samples are scarce.
Phage Therapy Potential	The gut phage catalogue documents Crassvirales lytic phages targeting Prevotella/Bacteroides with high confidence — candidates for phage therapy targeting IBD-associated dysbiosis.

9. Methods Summary

Block	Key Tools	Statistical Framework	ML Methods
Block 1	Python, scipy, scikit-learn	Mann-Whitney U, Wilcoxon, ICC	PAM clustering, LME
Block 2	sklearn, pandas, SHAP	Chi-square, Fisher exact, Cochran-Armitage trend	Random Forest, GBT, Logistic
Block 3	sklearn, PyTorch, TPOT	5-fold stratified CV, Youden's J	RF, GBT, Transfer Learning, AutoML
Block 4	Kraken2, Bracken, MetaBAT2, HUMAnN3	DESeq2-style normalisation, FDR correction	Unsupervised binning
Bonus	BioPython, DESeq2-style	Wald test, BH correction	PCA, hierarchical clustering

10. Datasets & References

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